EDA REPORT

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Executive Summary

This report presents an exploratory analysis of crime data across four UK police forces – **city of london, west midlands, essex and surrey**. The purpose of this to provide Nadine an overview of crime patterns in different regions in order to identify locations that may be desirable for residential property sales

The analysis examined crime types, regional crime rates per 1,000 population, monthly crime trends, geographic crime hotspots, and crime investigation outcomes. The results show that the City of London experiences significantly higher crime levels than the other regions, largely due to its dense commercial environment and high levels of daily commuter and tourist activity. In contrast, Surrey consistently records the lowest crime levels, with stable trends and very few crime hotspots, suggesting a safer and more desirable residential environment. Essex and the West Midlands display moderate crime levels, with incidents often concentrated in urban centres and commercial areas.

Based on these findings, Surrey and the West Midlands have been selected for further investigation. These two regions provide contrasting crime environments that will allow deeper analysis of neighbourhood-level crime patterns and their potential impact on property desirability and sales strategy

Introduction

This EDA exploratory analysis examines the different crime patterns across **four** UK police forces covering a span of 2 years. The goal was to help the real estate company assessing crime risk to understand the most/least desirable locations and support sales decisions, and recommend two police forces for deeper, later-stage analysis. Crime levels play a important role in determining desirability of residential areas. For real estate companies, understanding this will help inform decisions around property investment, pricing strategies and marketing approaches.

Data and methodology

The data was downloaded from the Data Police (Appendix A) containing crime category, location, month and outcome of the crime from the four police forces: **West-midlands, Essex, Surrey, City of London.**

The data covers January 2024 – January 2026

A population data per force was used to see the overall population of the forces.

Due to the population heavily differing amongst the 4 forces, raw crime counts will only favour the large forces, so to ensure a fair comparison was made, Crime rate per 1,000 population was used :

Crime rate per 1000 population = Crimes / population * 1000

To ensure Nadine can compare areas of risk based on a fair count.

For this exploratory analysis, the focus was placed on:

- Crime type distribution
- Regional comparisons
- Crime rate normalisation
- Identification of high- frequency crime categories
- Trends across regions

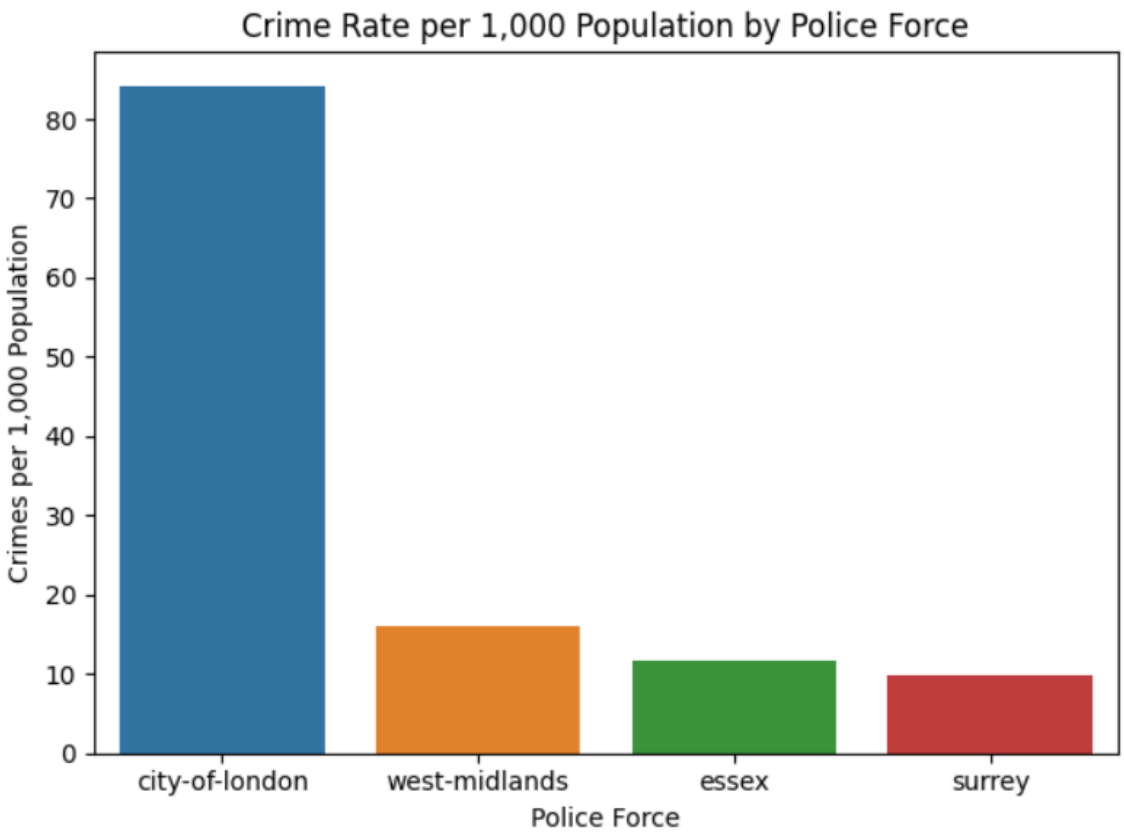
General comparisons between forces

	force	Crimes	population	Crime_rate_per_1000
0	west-midlands	677300	3036605	223.045144
1	essex	316905	1929610	164.232669
2	surrey	172999	1248649	138.548944
3	city-of-london	17820	15111	1179.273377

At a Glance, West Midlands appears to record the highest number of recorded crimes, followed by Essex, Surrey and City of London, but without accounting the total population it becomes misleading.

Crime rate per 1000 population distribution

By merging the forces, crimes, population and crime rate per 1,000 population it provides a fairer comparison as some forces are evidently smaller.

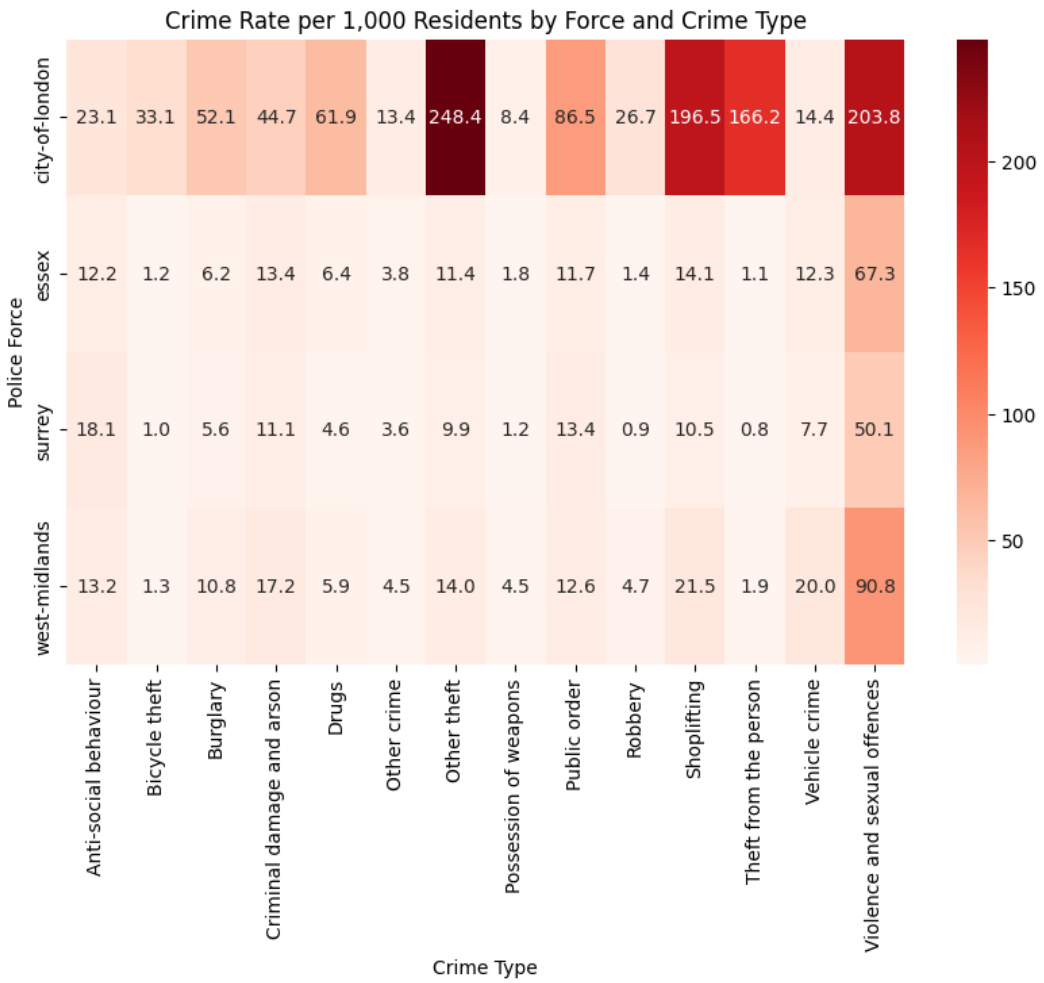


The initial analysis examined the total number of recorded crimes for each police force over the 2 years to understand how crime volumes vary between regions. From this chart we can see

despite having the smallest population size, city of London has the most crime rate in comparison to West-midlands despite it having the largest population. Amongst this ,Surrey is a low-risk area followed by Essex. This suggests Surrey is more attractive from a low crime risk-perspective, whilst West midlands may require careful positioning due to having the second highest-crime rate yet generally much lower in comparison to city of London.

After establishing overall crime rate per 1000 population, the next step is to understand which crime type drive the differences amongst the forces.

Crime rate heatmap



The heat map shows which crime types are high in each force which allows Nadine to identify each regions high crime types and helps highlight which areas are desirable, where the darker colours indicate higher crime rates, allowing patterns to be quickly identified. From this map, City of London immediately stands out showing extremely high rates of other theft, shop lifting, theft from the other person and violence of sexual offences which are often associated with very busy commercial environments where there are a lot of visitors, workers and tourists making this a opportunity for theft-related offences, then all the other forces.

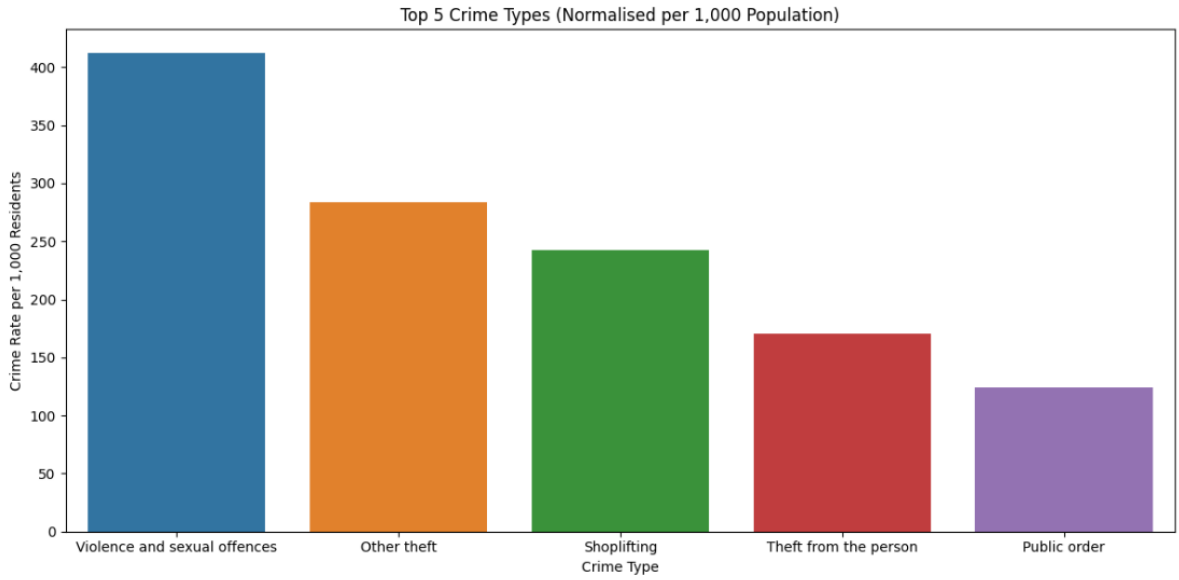
Surrey shows the lowest crime rates meaning it is more attractive for buyers as it shows a safer residential environment which can positively influence property desirability. Essex shows

moderate crime levels West midlands shows higher crime rates than Essex and Surrey particularly in violence and sexual offences and Shop lifting which may be influenced by larger urban centres like Birmingham which may experience higher crime levels than the smaller regions.

From a property market perspective, crime patterns can influence how attractive certain locations appear to buyers and so from this low crime regions like surrey may be more attractive for investment and higher crime urban areas like west midlands and city of London may not be.

Whilst the heatmap shows variation across all the crime types it may be more useful for the state estate to focus on the most common crime types overall to better understand the crimes that dominate between the four forces.

Most Common Crime Types

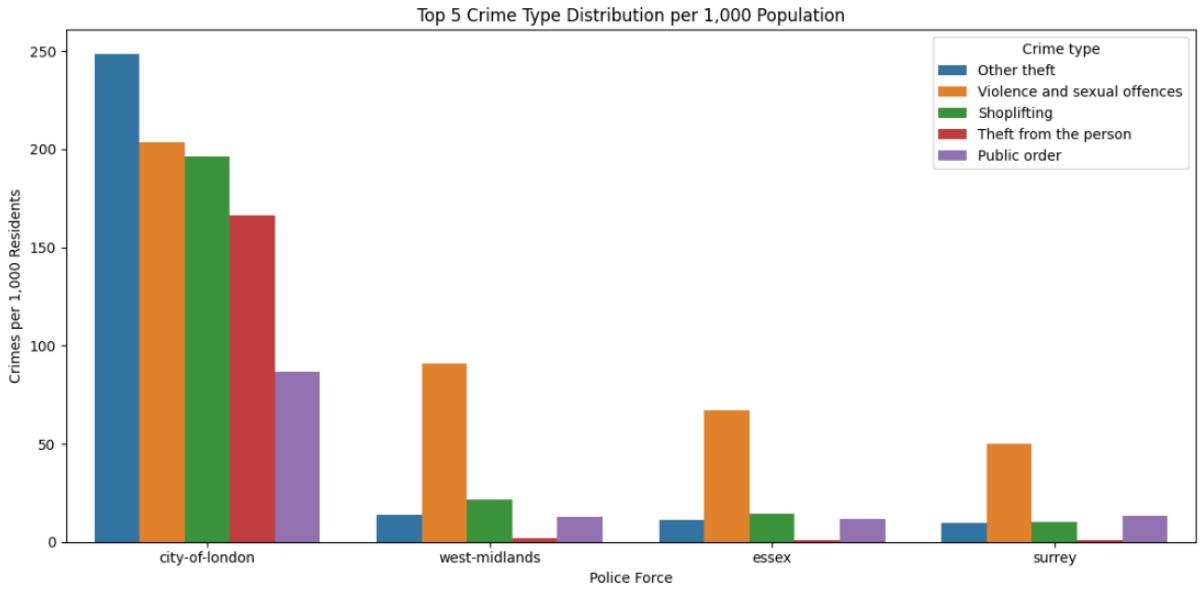


This shows the top 5 most common crime types across the selected police forces normalised per 1,000 people, which helps determine which types of crimes occur most frequently regardless of location. The analysis shows that the most common crime categories are :

1. Violence and sexual offences
2. Other theft
3. Shoplifting
4. Theft from the person
5. Public order offences

Violence and sexual offences appear more frequently than the other categories highlighting the importance of considering violent crime trends when assessing regional safety however these may vary between the police forces therefore to get a clearer view of the impacts within the forces the next analysis compares these top crime categories across each region.

Top Crime Types by Police Forces



To further understand regional differences, the top 5 crimes were then compared across the forces where it highlights how crime patterns differ between the regions and identifies which crimes contribute most to each area's overall crime levels.

From this it appears City of London shows extremely high rates of other theft, Shoplifting, theft from the person and Violence and sexual offences. Majority of the crime types are theft which is reflected from the high amount of businesses, tourism and retail activity within the popular city which may be leading to increase theft-related crimes.

West midlands shows the second highest with violence and sexual offences dominating in the category followed by shoplifting and other theft however it may be influenced by larger urban areas within the region so more detailed neighbourhood analysis may be needed to identify the safer areas within the big region.

Surrey consistently records the lowest crime rates across all top categories which suggests it may be one of the safer residential areas amongst the other regions, lower crime levels makes this area more attractive for families and long-term owners.

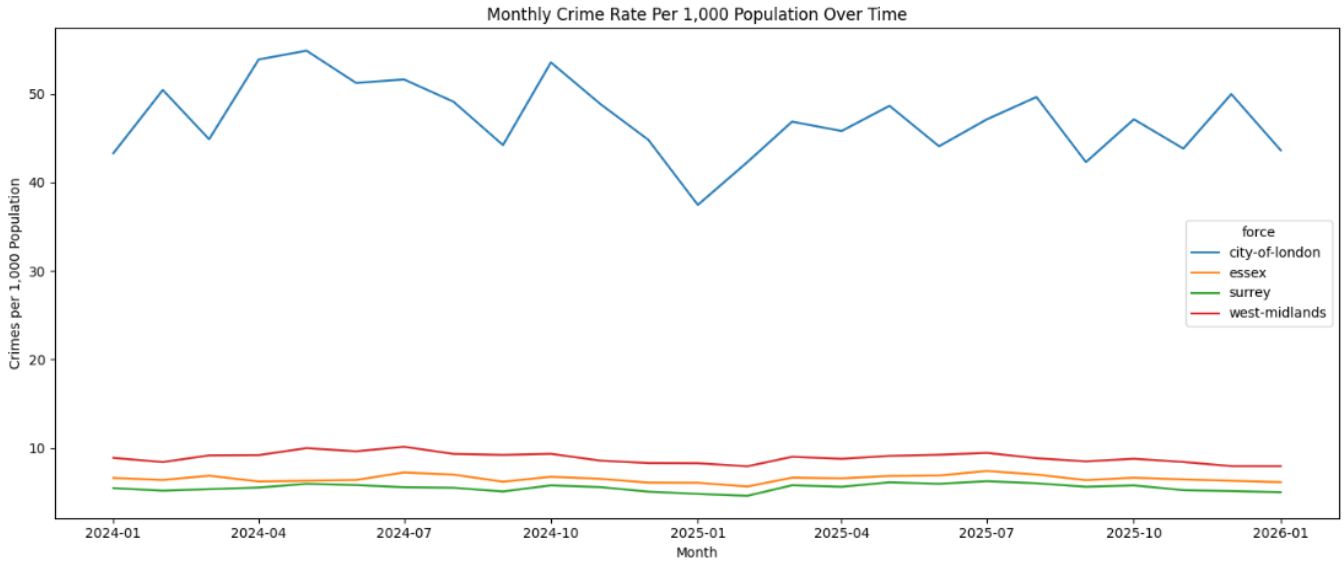
Essex shows moderate crime levels reinforcing the idea that it is a balanced mid risk region.

Monthly Crime Trends Across Police Forces

To further support decision making for Nadine, crime trends were analysed over time monthly to identify crime patterns for any seasonal fluctuations, unusual spikes and overall stability in crime levels across the four regions.

By understanding these trends with areas that are more stable with low crime levels, it is more attractive for residential buyers whilst regions with large fluctuations and frequent spikes may indicate a higher risk.

Crime Trends Over Time



The chart above shows the monthly crime rate per 1,000 people from January 2024 to January 2026 for the four forces.

City of London immediately stands out showing significant fluctuations in crime levels throughout the year, with several noticeable spikes and drops. There a lot of noticeable spikes between January 2024 to march to then dropping April but then suddenly sharpening towards Summer 2024 which is a repetitive pattern again during summer 2025 which could highlight the increased periods of higher tourism, increased retail activity and Seasonal events such as The Euros occurring in 2024 and declines between autumn of 2024 and autumn of 2025 which could be when tourism is low. Although City of London records high and fluctuating crime levels, these patterns are typical of a busy commercial place as it not a pure residential area so as a result crime data in this region should be interpreted with caution when assessing residential desirability.

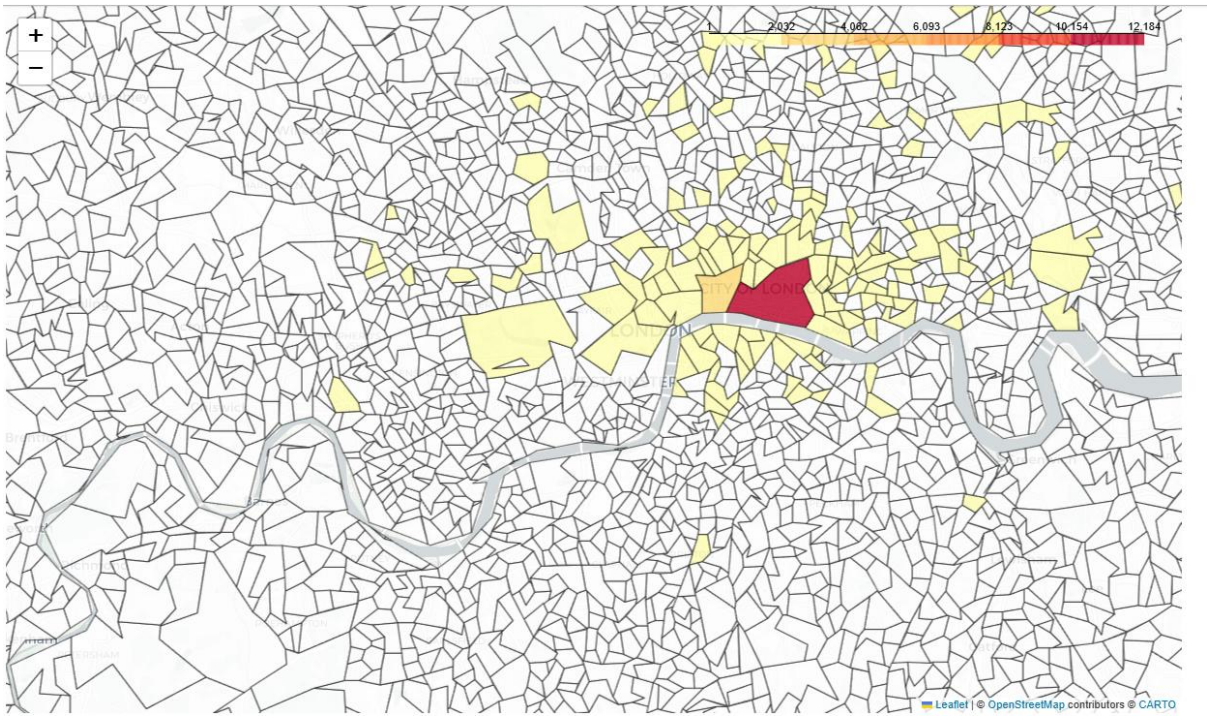
West Midlands shows the second highest crime rates with however moderate variation across months where the fluctuations are much smaller compared to the city of London. Crime levels remain relatively stable with no major spikes or declines. The consistent crime patterns can make it easier to identify specific neighbourhoods with stronger residential potential within the region.

Essex and Surrey show smaller crime rates population with minor fluctuations that appear relatively stable overtime which shows a more consistent crime environment which makes the area more predictable and reassuring for property buyers that want to relocate. From a real estate perspective, Surrey appears to be more desirable as it has the least number of crimes with a stable crime trend which provides more predictable property market conditions that will be more attractive for buyers as it is safer as stability is a key factor in buyer confidence.

Geographic Crime Distribution

To provide Nadine Green with a clearer understanding of where crime is heavily concentrated geographically, crime incidents were visualised to allow identifications of crime hotspots , which are specific areas where crime occurs more frequently. This is valuable for real estate decision-making as property desirability is influenced by neighbourhood-level safety as well rather than overall regional crime averages.

Crime Hotspots in the City of London



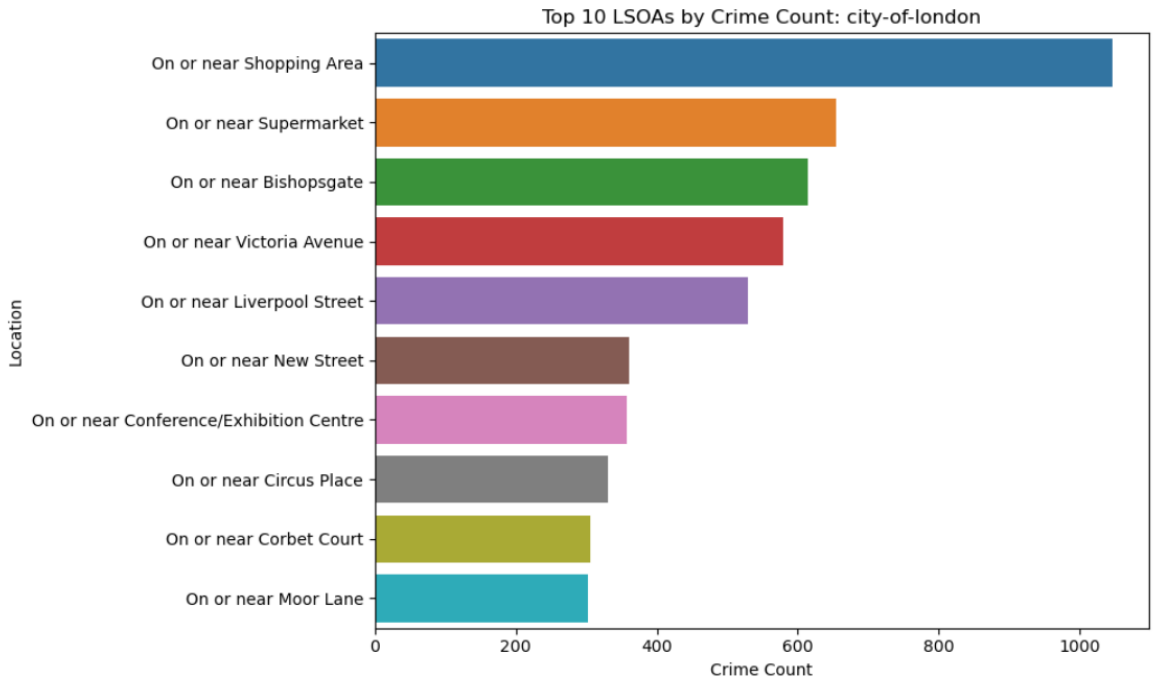
This zoomable map (on Jupiter notebook) shows areas within the City of London where crime incidents occur more often where the darker regions indicate higher crime densities, whilst lighter areas represent locations with fewer recorded incidents.

The map shows a clear amount of crime in central commercial districts particularly the central which area areas with high levels of business activity and tourism, therefore this pattern is expected as it attracts large amounts of commuters, shoppers and visitors each day.

Crime in the city of London is therefore high in busy commercial zones rather than the residential neighbourhoods which are outside the city therefore the density observed in these areas may not necessarily reflect the safety of nearby residential locations, however properties within these high concentrated areas face challenges related to safety and security.

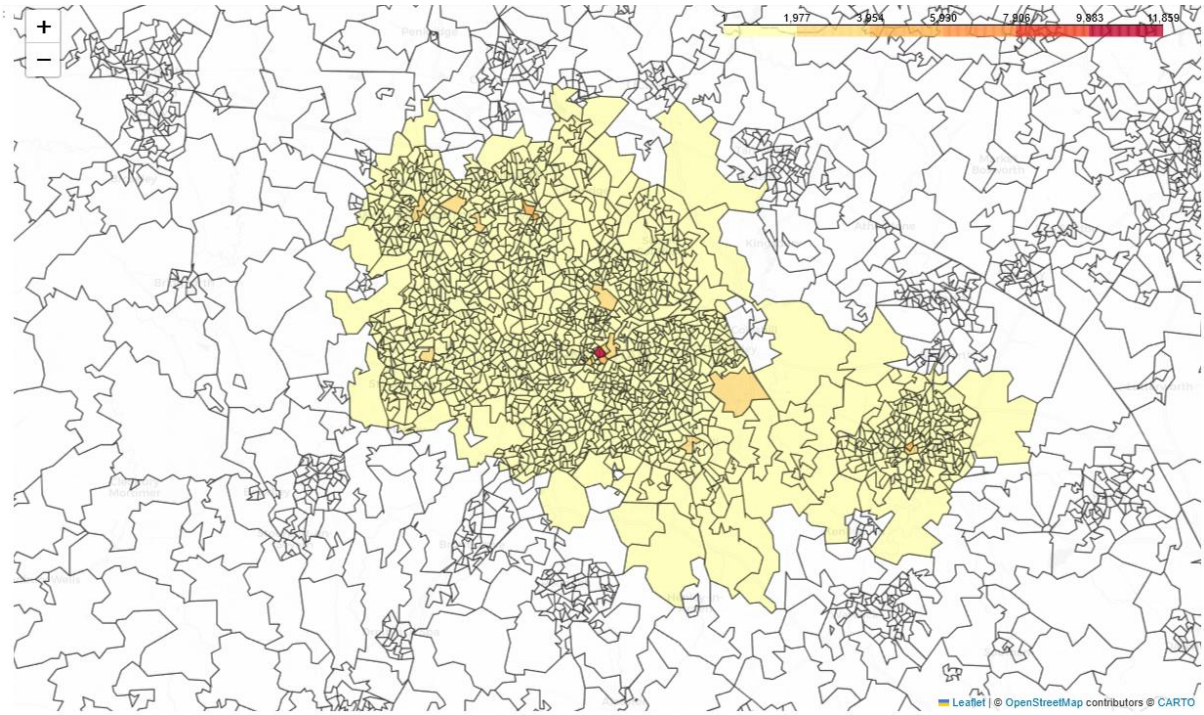
Crime Locations in the City of London

The bar chart shows the Top 10 locations by crime count to further highlight where crimes occur within the city of London.



The highest crime locations include shopping areas, near supermarkets and near busy tourist locations near Bishopsgate, Victoria avenue and Liverpool Street which contains transport facilities meaning high levels of public activity. This highlights the high shop lifting category and violent offences city-of-London currently dominates in meaning Residential properties near high retail or transport hubs may experience high amount of crime levels, however these areas also benefit from strong commercial value and accessibility which can still attract certain buyers.

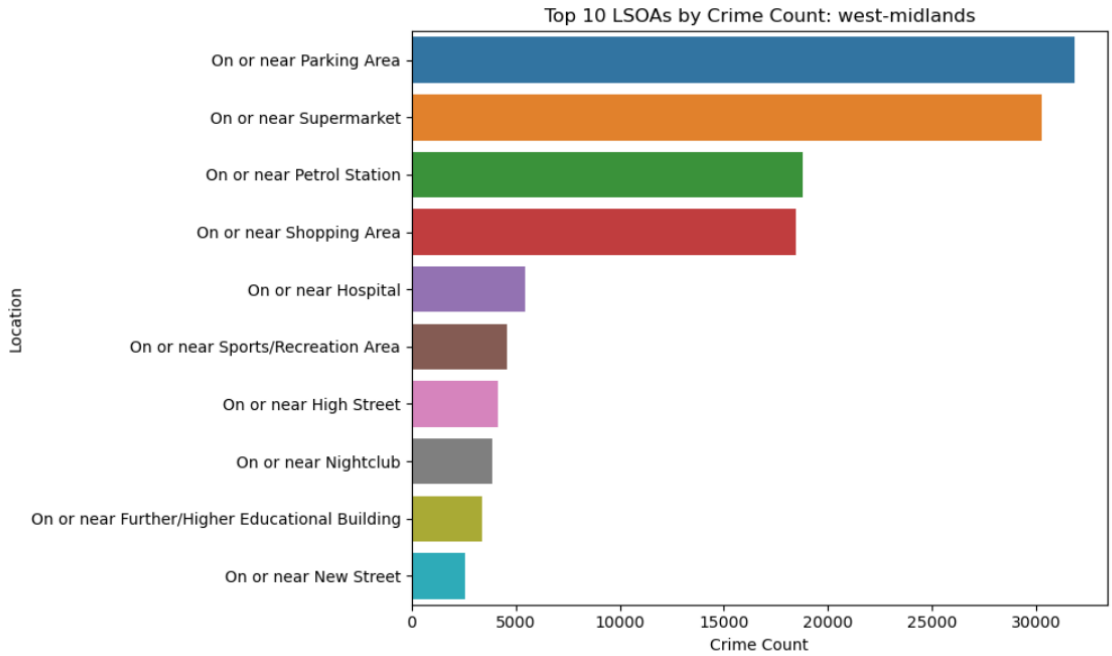
Crime Hotspots in west midlands



This map shows a clear patterns where the deep red areas are clustered around Birmingham which reflects the regions population density and the among of commercial, retail and transport modes, and as you move outward from the centre the colour intensity gets lighter meaning the crime opportunities fall where activity is at its highest, therefore this makes the relationship visible by showing how crime is drawn towards the busiest areas of the region showing the rural areas experience far lower levels of crime.

Crime Locations in west midlands

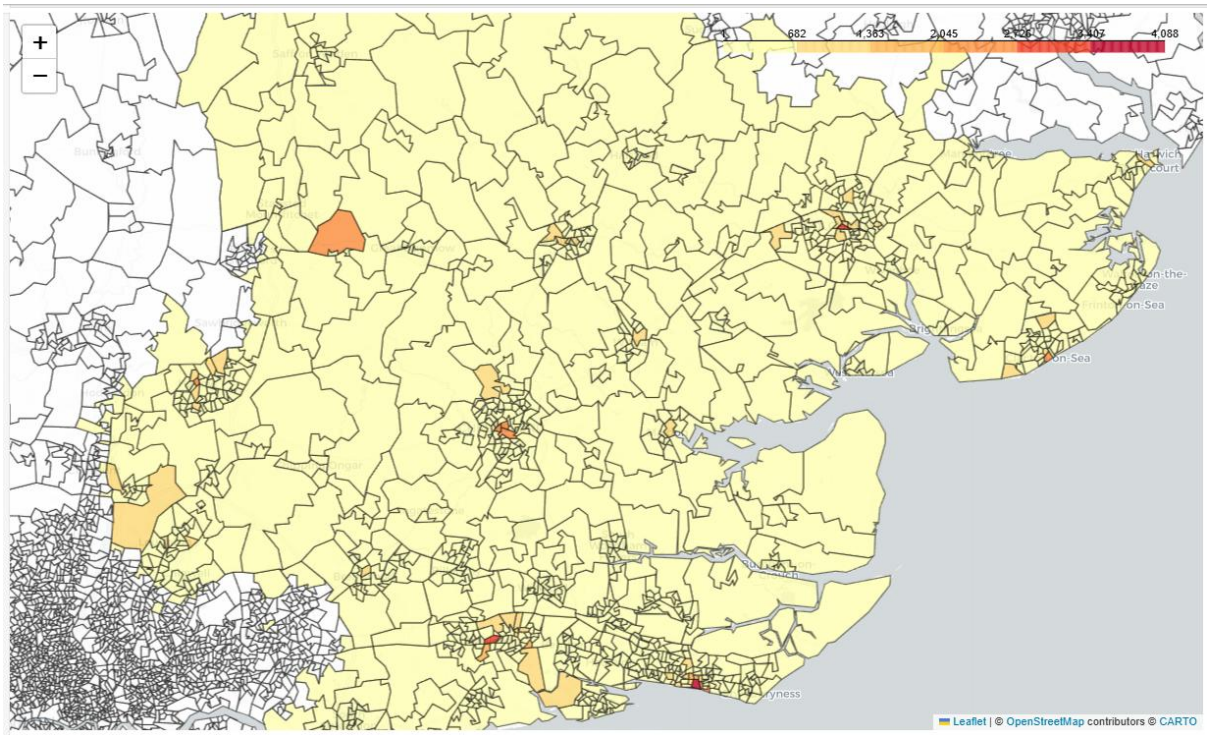
A similar analysis was done for the west midlands to identify areas where crime incidents are happening more frequently,



The results show that crime frequency occurs near parking areas, supermarkets, petrol stations and near shopping areas. In comparison to the city of london, these reflect a mix of commercial and venues that require parking's. This chart suggests that incidents often occur in public access locations where larger numbers of people gather or travel through which contributes to shoplifting and violences offences which dominates crime types in west midlands.

From this analysis it shows crime hotspots are commonly located near retail areas again, and transport hubs as well. Although this may appear less desirable from a crime perspective, it remains high value locations due to strong economic activity and accessibility and overall Residential neighbourhoods within the same police force area experience lower crime levels than nearby commercial centres.

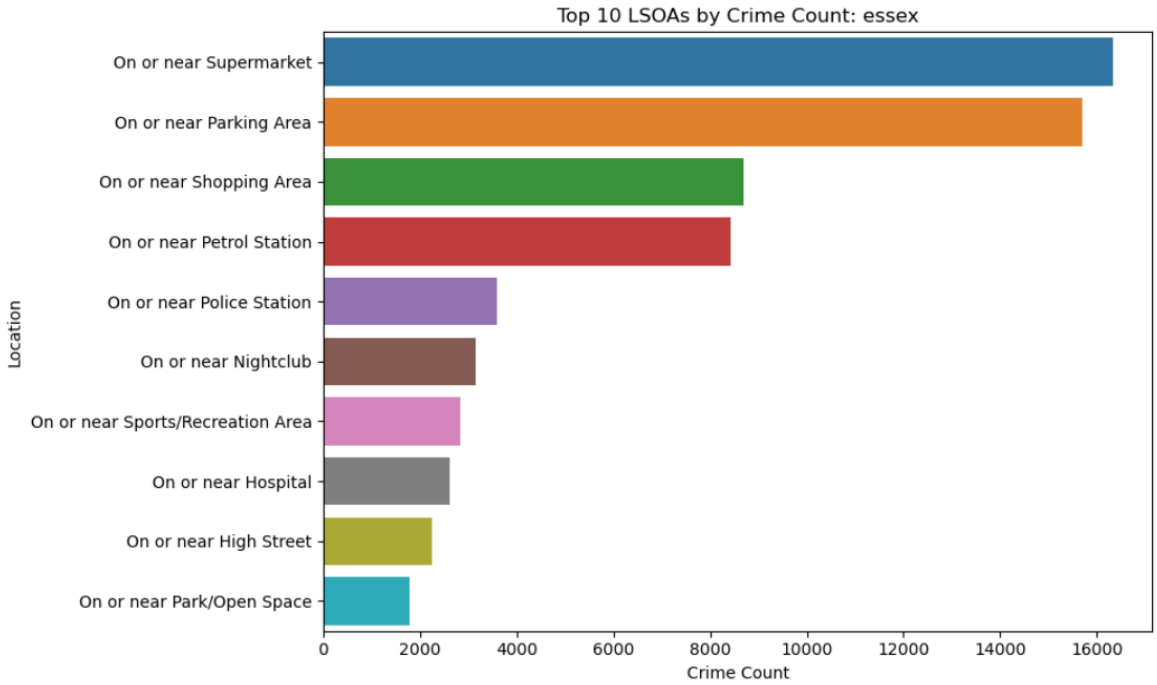
Crime Hotspots in Essex



The Essex map shows a multi crime pattern, with several distinct hotspots spread across the county rather than one single dominant hotspot as seen by city of london and west midlands. This reflects Essex's population being large and a mixed urban rural county with multiple major towns than one central city with darker clusters around Southend-on-Sea, Basildon, Colchester by zooming into the darker clusters in the notebook analysis, These areas have the highest population density, busiest commercial centres and major transport links which means it will generally have more crimes. The orange areas represent moderate crime levels with light yellow with the lowest amount of crimes.

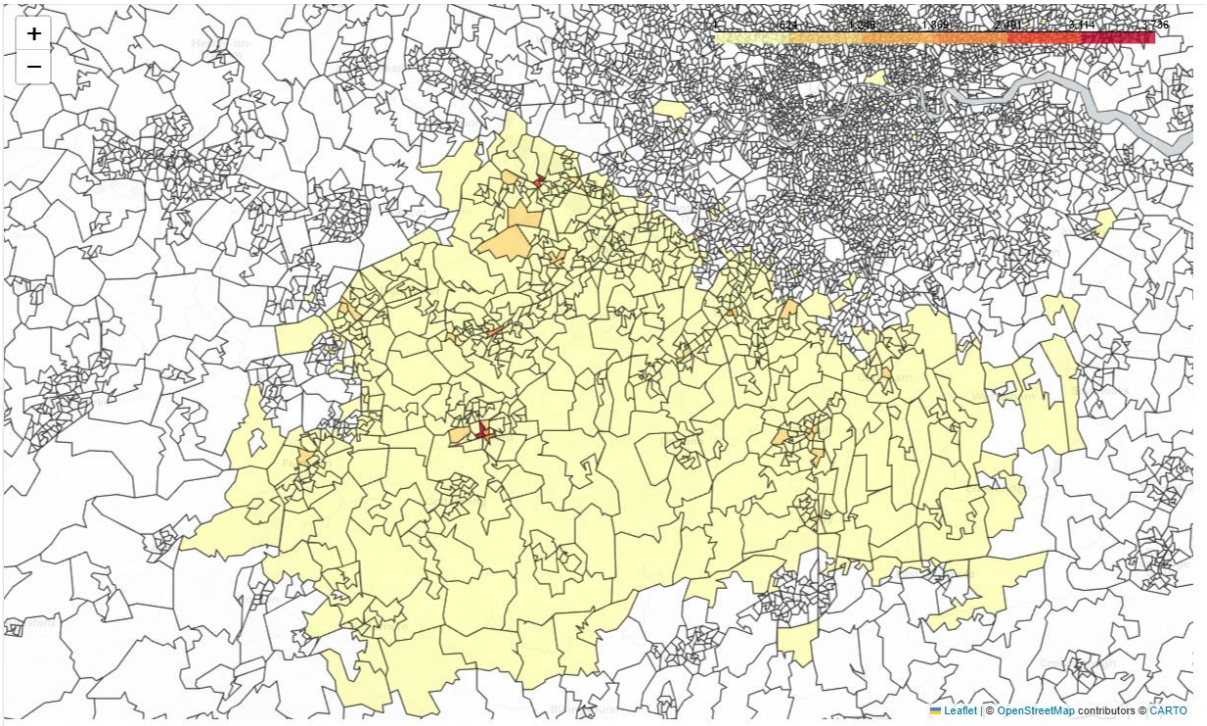
From this the Essex map sits between the extremes shown by other forces as it is less concentrated than the city of London which has one single large hotspot. It is less dominated by one core than the west midlands , where Birmingham drives the entire pattern.

Crime Locations in Essex



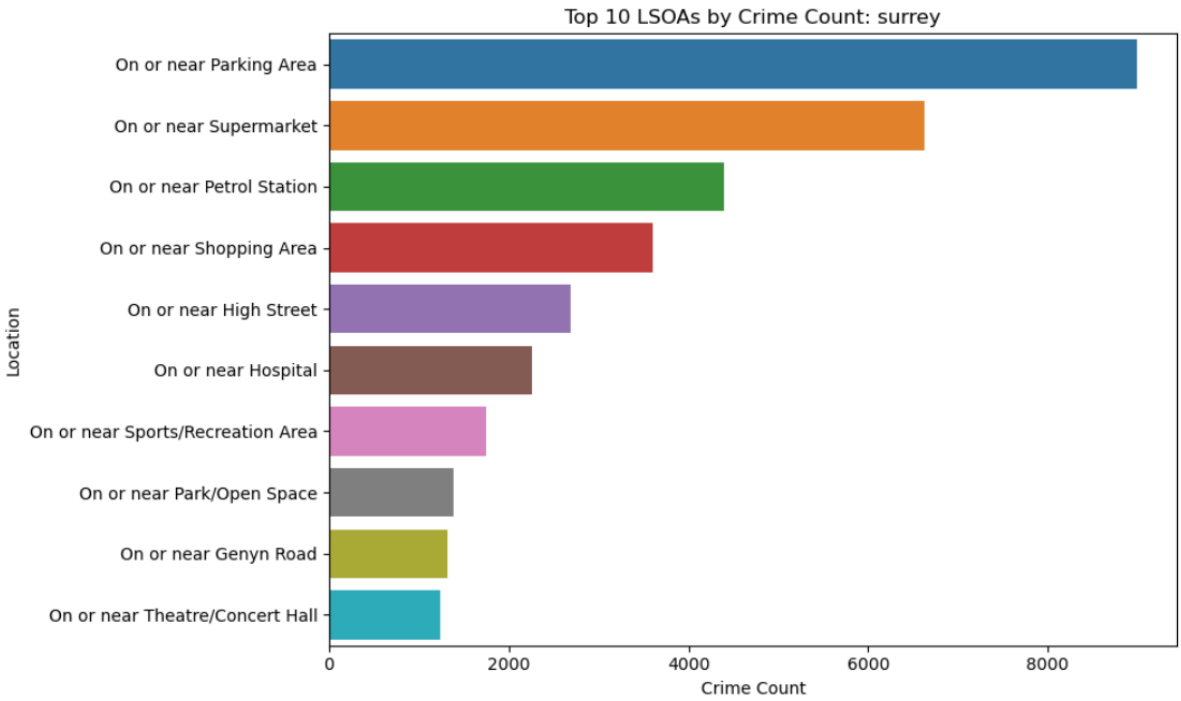
The next chart adds further insights derived from the map where it explains why those hotspots exist by revealing the specific environments that generate the highest volumes of crime, where the top locations in Essex include near supermarkets, parking areas, shopping areas and petrol stations which are hight activity environments and so will naturally attract more crime due to having a large number of people, frequent movements, retail and public gatherings which align with major towns like southend on sea with a lot of parking and public big fair ground and beaches, as well as Chelmsford, Basildon and Colchester. The combination of these charts show that the hotspots are not random but with high activity locations.

Crime Hotspots in Surrey



The surrey map shows different hotspot amounts in comparison to the other forces, it forms patterns of isolated but intense hotspots surrounded by large areas of low-crime levels which reflects how there are less busy town centres and large amounts of low-crime rate residential neighbourhoods. The darker hotspots correspond to the commercial and transport modes such as Epsom and Guildford where it contains high streets and shopping centres. Overall compared with the other police forces, Surrey demonstrates the lowest concentration of crime overall as most of the region appears in lighter colours indicating lower crime rates and fewer reported incidents across the majority of the neighbourhoods.

Crime Locations in Surrey



The top 10 locations for Surreys Crime count shows they are usually near Parking areas, Supermarkets, petrol stations, shopping areas, which is exactly why the maps were able to highlight the hotspots that contributed to high activity environment, however the overall crime count is much lower than those observed in other regions.

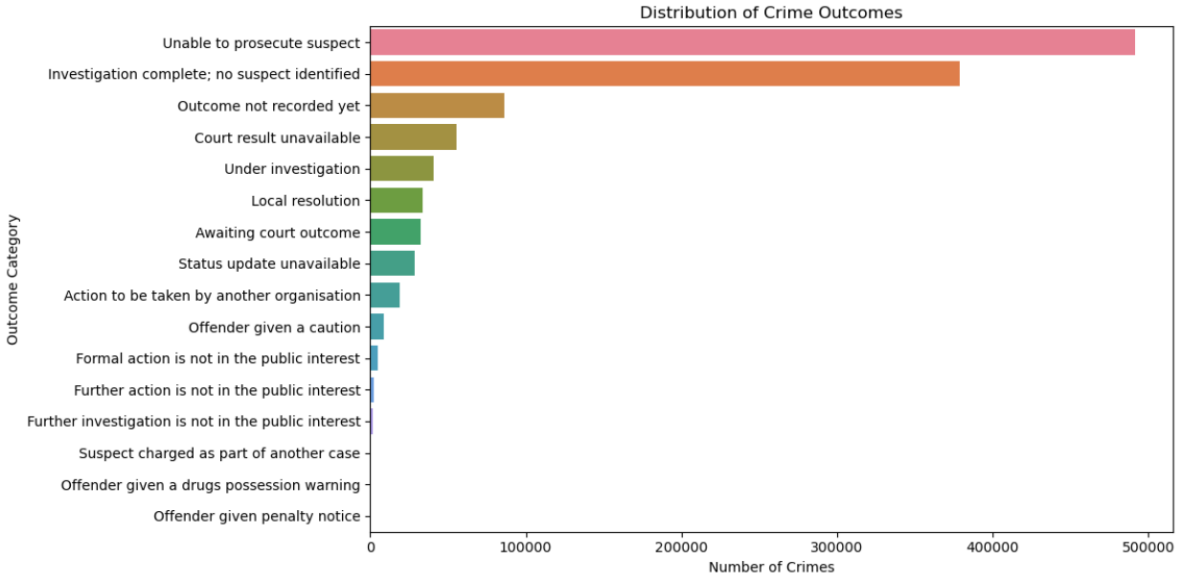
Therefore for a real estate perspective , Surrey appears to be one of the most desirable regions amongst the others analysed as it represents a relatively safe and stable with crime limited to small commercial activity areas.

Crime Outcomes Distribution

In addition to analysing crime types and locations, it is also important to investigate the crime outcomes to describe the result of a police investigation after a crime has been recorded. Understanding these outcomes will provide insights to how crimes are resolved which can influence public perceptions of safety and trust in the areas.

The following analysis will explore the overall distribution of crime outcomes across the selected police forces.

Overall crime outcome distribution



The distribution shows that most of the recorded crimes fall into unable to prosecute suspect, investigation complete – no suspect identified, outcome not yet recorded, court result unavailable and still under investigation.

The most common outcome shows that although a suspect may have been identified there may not be enough evidence to proceed with prosecution.

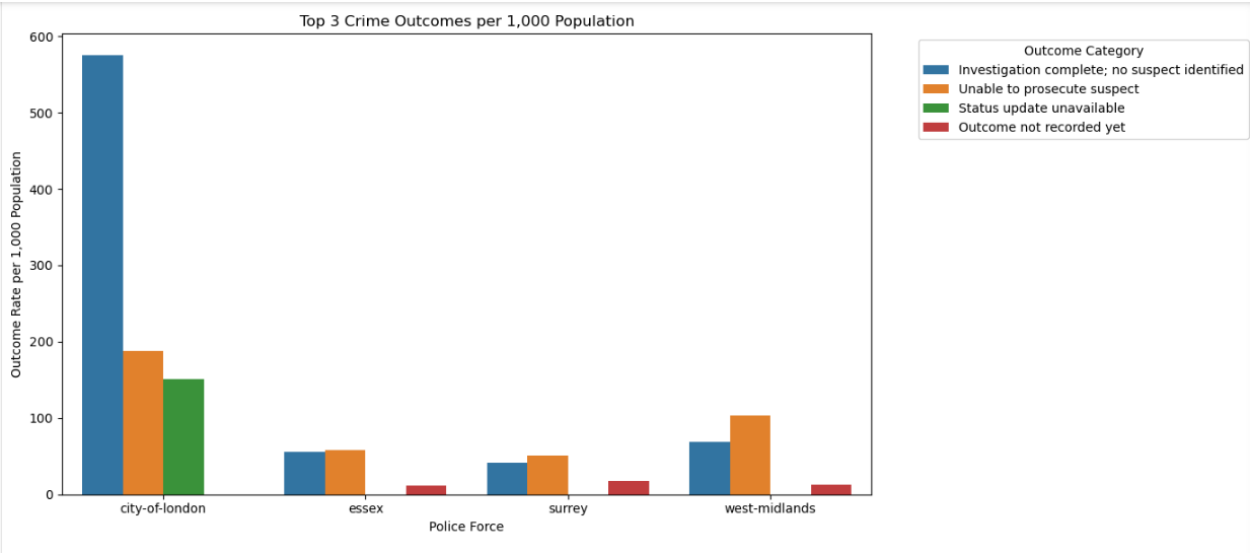
The second most frequent outcome suggests that many crimes occur where offenders are not easily traceable which are common for offences such as theft or shoplifting.

This distribution suggests that most of the crimes don't result in sanctions as suspects are either not identified or prosecution isn't possible which is important to assess for safety and attractiveness of a location.

This finding aligns with earlier results showing that theft-related crimes and incidents in busy commercial areas are among the most common crime types

Top Crime Outcomes Per 1,000 Population by Police Force

To further understand how crime investigations progress across different regions, the most common crime outcomes were analysed per 1,000 people for each police force to allow for a fair comparison between areas with different population sizes.



From this chart its clear that The City of London has the greatest number of crimes with investigations completed with no suspects identified followed by unable to prosecute suspect and status unavailable. These high values are consistent with earlier findings that London experiences significantly higher crimes overall where majority occurs in busy public areas such as shopping districts and transport areas where offenders may be difficult to identify.

West midlands shows moderate outcome rates higher than surrey and Essex but significantly lower that London where the most frequent outcome is unable to prosecute a suspect followed by investigation complete with no suspects identified and then outcome not recorded. These figures reflect the regions larger urban population and overall high crime volume which may lead to more complex investigations and cases where prosecution is difficult.

Essex shows much lower crime rates where its mostly common to unable to prosecute suspects followed by investigation complete and outcome not recorded the lowest which align with earlier findings that Essex has a moderate crime level overall which are around commercial locations such as shopping areas and parking facilities. Essex appears overall to have manageable crime levels and investigation outcomes.

Surrey records the lowest outcome rates amongst all forces which is consistent with the regions lower crime levels observed throughout the analysis, these lower values reflect the overall lower crime frequency in surrey as previously seen during crime type analysis, monthly trends and crime hotspot maps. The consistent lower crime and outcome rates reinforce the point that surrey represents one of the safest and most desirable residential regions amongst the four forces

Conclusion and Recommendation

This exploratory data analysis explored crime patterns across four forces using the crime data from UK police open data where the goal was to help provide Nadine green an overview of crime trends across the regions to identify locations that may be desirable for property sales and investment. Several key factors were analysed including total crime volumes, crime type distributions, crime rates per 1,000 population, monthly crime trends, spatial crime hotspots and crime investigation outcomes. The results show clear differences between the selected regions were the City of London consistently recorded the highest crime levels across multiple

analyses. Crime types such as theft, shoplifting, and violence were particularly prominent, and crime rates per 1,000 people were significantly higher than in the other police forces. Monthly crime trends also showed large fluctuations, with multiple spikes throughout the year. Spatial mapping further revealed strong crime concentration around busy commercial areas such as shopping districts, transport hubs, and financial centres. These patterns are likely influenced by the high level of daily commuters, tourism, and business activity in the area. While this does not necessarily reflect residential safety directly, it indicates that the City of London is a high-activity environment where crime is strongly associated with commercial density.

In contrast Surrey consistently demonstrated the lowest crime levels across all analyses. Crime rates per 1,000 people were significantly lower, and monthly crime trends remained stable with minimal spikes. Spatial analysis also showed very few crime hotspots, with incidents mostly occurring around commercial locations such as supermarkets, parking areas, and shopping centres. These findings suggest that Surrey provides a more stable and lower-risk environment, which may make it particularly attractive for residential buyers and families prioritising safety.

Essex showed moderate crime levels across most analyses, were crime hotspots identified included southend on sea, Chelmsford, Basildon, Colchester often around busy areas such as shopping centres however despite this crime levels remained significantly lower than city of london and trends appear relatively stable over time meaning essex could represent a balanced residential area.

West midlands also showed moderate crime patterns however higher in comparison to Essex and Surrey. The overall analyses showed that a higher proportion of cases where suspects not being prosecuted however these patterns reflect the regions larger population.

Based on these findings **two police forces** are recommended for further analysis in the next stage of the project:

- **Surrey Police**, representing a low-crime region with stable trends and high residential desirability
  Many areas are perceived safe due to this and so is great for positioning properties outside hotspots . It is a strong contrast between quiet residential areas and busy town centres.
- **West Midlands Police**
  This stands out as a force where it is a large, active housing market with strong demand but also clear safety concerns in some neighbourhoods. Crime is highly concentrated in central , busy areas such as city centre, shopping areas, parking however many suburban areas show much lower crime levels

By comparing these two contrasting regions in greater detail further analysis can provide more valuable insights into housing demands and potential sales strategies as this analysis gives a foundation of a high – level understanding of crime trends for more detailed investigation in the next stage of the project.

Appendix

Appendix A

Crime data

Data downloads | data.police.uk

Population data

Population estimates for police force areas in England and Wales by single year of age and sex, mid-1991 to mid-2024 - Office for National Statistics